DEPARTAMENTO DE ELETRÓNICA, TELECOMUNICAÇÕES
E INFORMÁTICA

LICENCIATURA EM ENGENHARIA DE COMPUTADORES E INFORMÁTICA

ANO 2025/2026

MÉTODOS PROBABILÍSTICOS PARA ENGENHARIA DE COMPUTADORES E INFORMÁTICA

PRACTICAL GUIDE NO. 8

1. Consider a link for IP communications from one router to another router with a capacity of 100 Mbps (1 Mbps = 10^6 bps). The link supports one flow of IP packets whose throughput is 90 Mbps and the IP packet sizes are exponentially distributed with an average of 800 Bytes. The link has an associated queue of size Q (in number of packets).

(a) Compute the average packet delay (in milliseconds) and the packet loss rate (in %) suffered by the flow when the queue size is Q = 10 packets. Results:

```
Avg. packet delay  = 0.318 msec.
Packet loss rate   = 4.373%
```

(b) Compute the minimum queue size Q such that the packet loss rate suffered by the flow is not higher than 0.01%. Results:

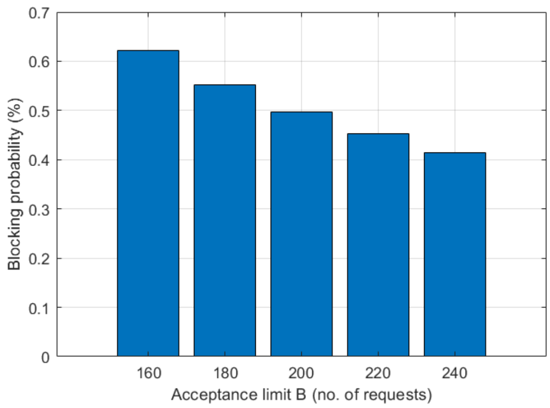
```
Minimum queue size = 65 packets
Packet loss rate   = 0.00956%
```

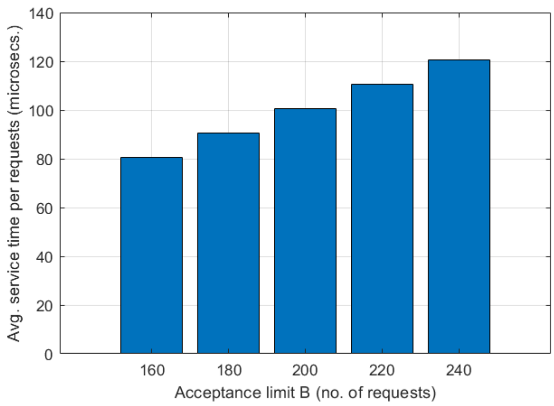
2. Consider a data service server such that, when the server is attending 200 simultaneous requests, it serves each request with an exponentially distributed time of 200 µsecs on average (1 µsecs = 10^{-6} seconds). The server receives service requests at an average rate λ (in requests/sec) and has an acceptance limit of B simultaneous requests.

(a) Compute the service blocking probability and the average service time per request when $\lambda = 1 \times 10^6$ and B = 200. Results:

```
Service blocking probability  = 0.49751%
Avg. service time per request = 100.50 microseconds
```

(b) Plot the service blocking probability (in %) in one bar chart and the average service time per request (in µsecs) in another bar chart considering B = 160, 180, 200, 220 and 240, when $\lambda = 1 \times 10^6$. What do you conclude? Results:





3. Consider a *Call Center* service with X operators such that each operator attends each client with an exponentially distributed time of 5 minutes. The service receives requests at an average rate λ (in requests/hour).

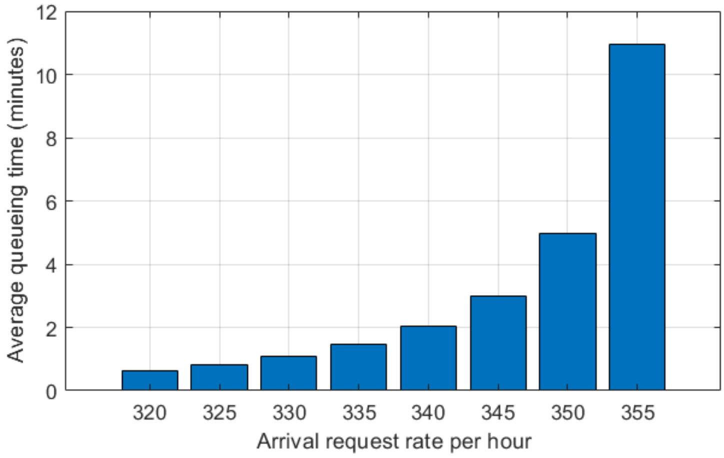
(a) Compute the probability of all operators being occupied and the average queuing time per request when $\lambda = 350$ and X = 30.

```
Prob. of all operators occupied = 82.799%
Avg. queuing time per request   = 298.075 seconds
```

(b) When $\lambda = 350$, compute the minimum number X of operators required so that the average queuing time per request is not higher than 15 seconds. For the value of X found, compute the probability of all operators being occupied and the average queuing time per request. Results:

```
Minimum number of operators     = 35
Prob. of all operators occupied = 21.765%
Avg. queuing time per request   = 11.194 seconds
```

(c) Plot the average queuing time per request considering λ = 320, 325, 330, 335, 340, 345, 350 and 355, when the number of operators is X = 30. What do you conclude? Results:



4. Consider a server based on a VM (Virtual Machine) provided by a Hypervisor that runs directly over the hardware of a physical machine. The average time between failures is 2 years (for hardware failures), 180 days (for Hypervisor failures) and 90 days (for VM failures). Hardware failures are solved in two days (on average) but do not include software recovery. Hypervisor failures are solved in 12 hours (on average) and include VM recovery. VM failures are solved in 1 hour (on average). Assume that 1 year has 365 days.

(a) Define in Matlab the matrix of the state transition rates (in hour $^{-1}$) of the Markov chain that allows to compute the availability of the VM. Assume states 1 (available), 2 (repair of hardware failure), 3 (repair of hypervisor failure) and 4 (repair of VM failure). Result:

```
Matrix of the state transition rates:
      0         0    0.0833    1.0000
 0.0001         0         0         0
 0.0002    0.0208         0         0
 0.0005         0         0         0
```

(b) Determine the server availability. Result:

```
Server availability = 99.338%
```

(c) Determine the MTBF (mean time between failures) in days and the MTTR (mean time to repair) in hours of the server. Results:

```
MTBF (mean time between failures) = 55.44 days
MTTR (mean time to repair)        = 8.87 hours
```